

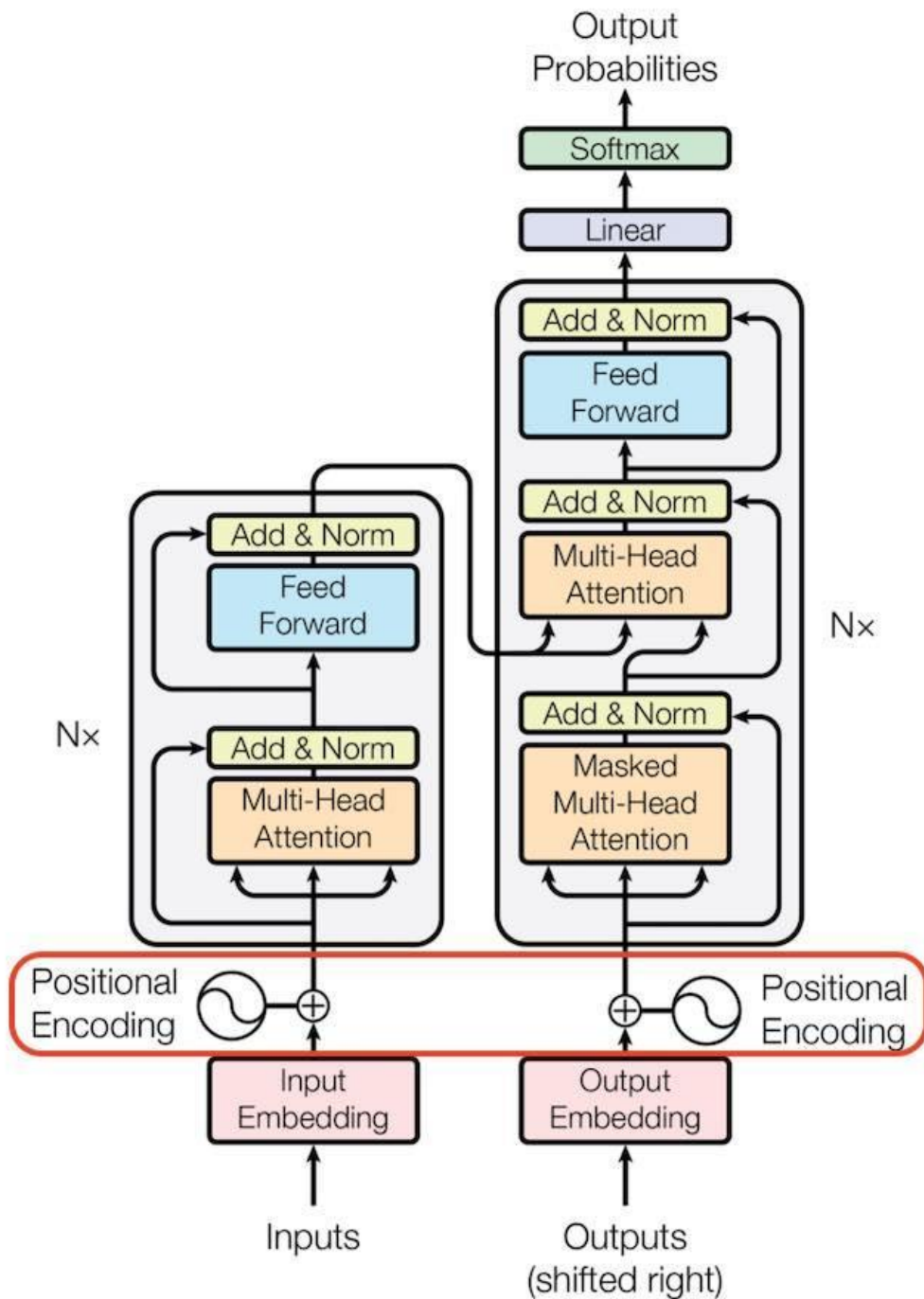


An Overview of Positional Embeddings

Agenda

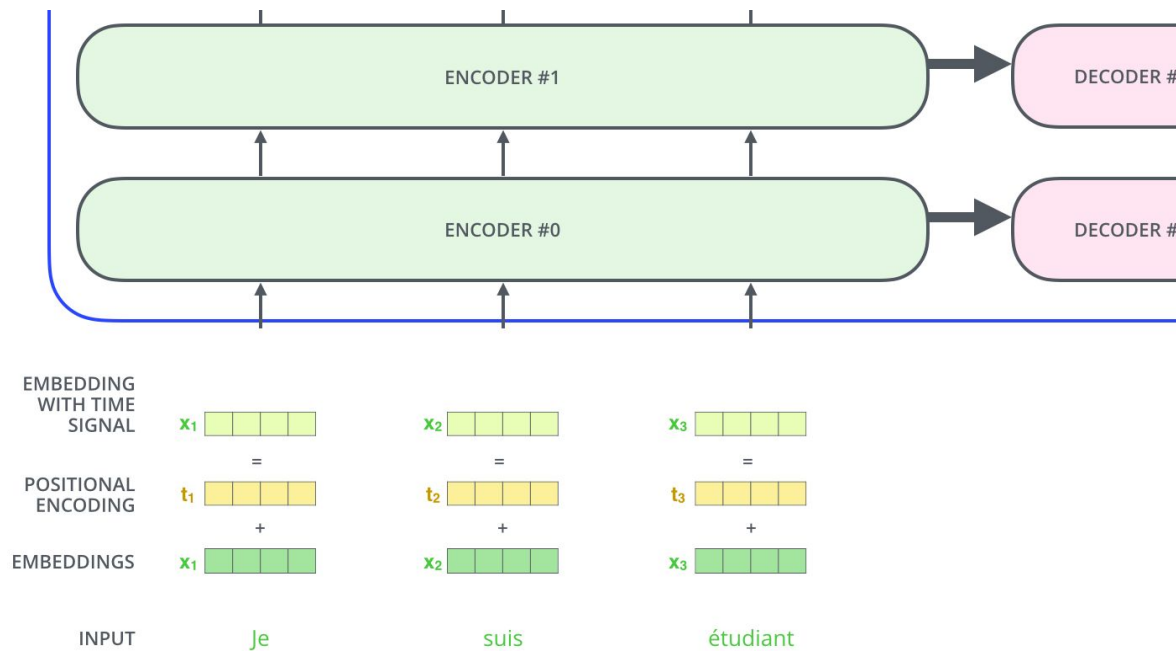
- **What is the Positional Embedding?**
- **Absolute Positional Embedding**
- **Relative Positional Embedding**
- **Rotary Positional Embedding**





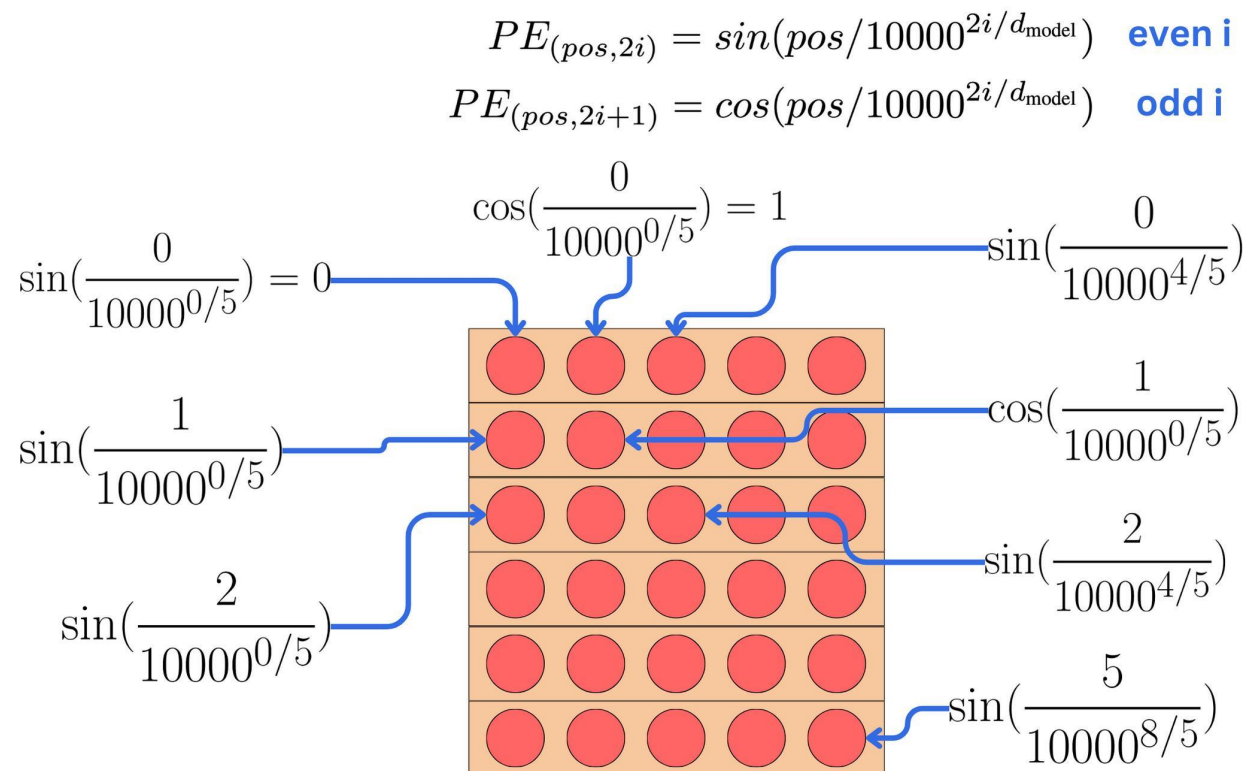
What is the Positional Embedding?

What is the Positional Embedding?



The positions of the words are encoded by a vector of the same size as the word embedding and are added directly to the word embedding.

Absolute Positional Embedding



Absolute Positional Encoding

Absolute Position Encoding in Original Transformer

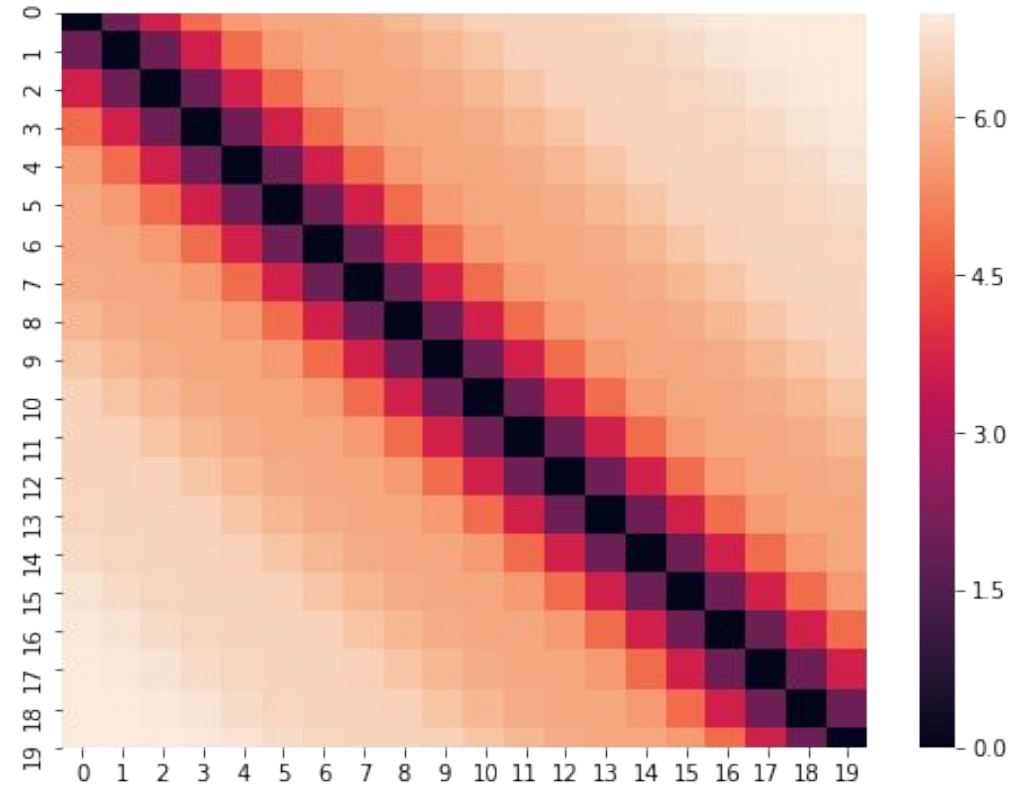
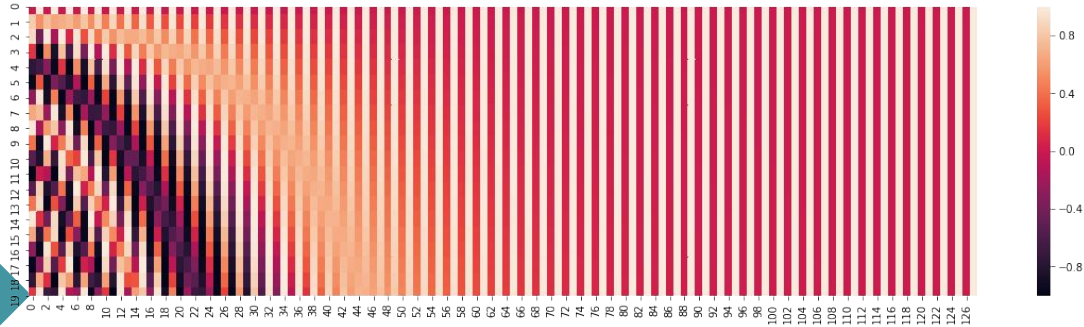
$$\begin{cases} PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \\ PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \end{cases}$$

- Where pos is the position of the word in the sentence, and i is the position in the positional encoding vector.

Absolute Positional Encoding

0	0	0	0	0	8	1	0	0	0
1	0	0	0	1	9	1	0	0	1
2	0	0	1	0	10	1	0	1	0
3	0	0	1	1	11	1	0	1	1
4	0	1	0	0	12	1	1	0	0
5	0	1	0	1	13	1	1	0	1
6	0	1	1	0	14	1	1	1	0
7	0	1	1	1	15	1	1	1	1

Absolute Positional Encoding



Properties

- Let N be the set of positions of words in the sentence, and R^D be the set of position embeddings.
- Let $\phi(\vec{x}, \vec{y})$ be a function that computes the similarity between two position $\vec{x}, \vec{y}$ ($\vec{x}, \vec{y} \in R^D$) corresponding to two words at positions x and y ($x, y \in N$).
- The function $\phi(\vec{x}, \vec{y})$ must satisfy the following properties:

1. **Monotonicity:**

$$\forall x, m, n \in N: m > n \Leftrightarrow \phi(\vec{x}, \overrightarrow{x+m}) < \phi(\vec{x}, \overrightarrow{x+n})$$

2. **Translation invariance:**

$$\forall x_1, x_2, \dots, x_n, m \in N: \phi(\overrightarrow{x_1}, \overrightarrow{x_1+m}) = \phi(\overrightarrow{x_2}, \overrightarrow{x_2+m}) = \dots = \phi(\overrightarrow{x_n}, \overrightarrow{x_n+m})$$

3. **Symmetry:**

$$\forall x, y \in N: \phi(\vec{x}, \vec{y}) = \phi(\vec{y}, \vec{x})$$

Properties

- If we choose the similarity function as the inner product, the sinusoidal function satisfies property 3.
- The sinusoidal function satisfies property 2 because there exists a matrix M such that:

$$M \cdot \begin{bmatrix} \sin(w_k \cdot t) \\ \cos(w_k \cdot t) \end{bmatrix} = \begin{bmatrix} \sin(w_k \cdot (t + \phi)) \\ \cos(w_k \cdot (t + \phi)) \end{bmatrix} \quad (1)$$

Properties

Proof:

- Let $M = \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}$, then:

$$(1) \Leftrightarrow \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix} \cdot \begin{bmatrix} \sin(w_k \cdot t) \\ \cos(w_k \cdot t) \end{bmatrix} = \begin{bmatrix} \sin(w_k \cdot (t + \phi)) \\ \cos(w_k \cdot (t + \phi)) \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix} \cdot \begin{bmatrix} \sin(w_k \cdot t) \\ \cos(w_k \cdot t) \end{bmatrix} = \begin{bmatrix} \sin(w_k \cdot t) \cos(w_k \cdot \phi) + \cos(w_k \cdot t) \sin(w_k \cdot \phi) \\ \cos(w_k \cdot t) \cos(w_k \cdot \phi) - \sin(w_k \cdot t) \sin(w_k \cdot \phi) \end{bmatrix}$$

- Choose $\begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix} = \begin{bmatrix} \cos(w_k \cdot \phi) & \sin(w_k \cdot \phi) \\ -\sin(w_k \cdot \phi) & \cos(w_k \cdot \phi) \end{bmatrix}$, then we have the Q.E.D.

Properties

Property 1:

- We have:

$$\phi(\vec{x}, \vec{y}) = \text{sum} \left(\begin{bmatrix} \sin(w_1 x) \\ \cos(w_1 x) \\ \vdots \\ \sin(w_{\frac{D}{2}} x) \\ \cos(w_{\frac{D}{2}} x) \end{bmatrix} \odot \begin{bmatrix} \sin(w_1 y) \\ \cos(w_1 y) \\ \vdots \\ \sin(w_{\frac{D}{2}} y) \\ \cos(w_{\frac{D}{2}} y) \end{bmatrix} \right) = \text{sum} \left(\begin{bmatrix} \sin(w_1 x) \sin(w_1 y) \\ \cos(w_1 x) \cos(w_1 y) \\ \vdots \\ \sin(w_{\frac{D}{2}} x) \sin(w_{\frac{D}{2}} y) \\ \cos(w_{\frac{D}{2}} x) \cos(w_{\frac{D}{2}} y) \end{bmatrix} \right) = \sum_{i=0}^{\frac{D}{2}} \cos(w_i (x - y))$$

- To examine Monotonicity, we need to analyze the monotonicity of the function $\psi(m) = \sum_{i=0}^{\frac{D}{2}} \cos(w_i m)$:

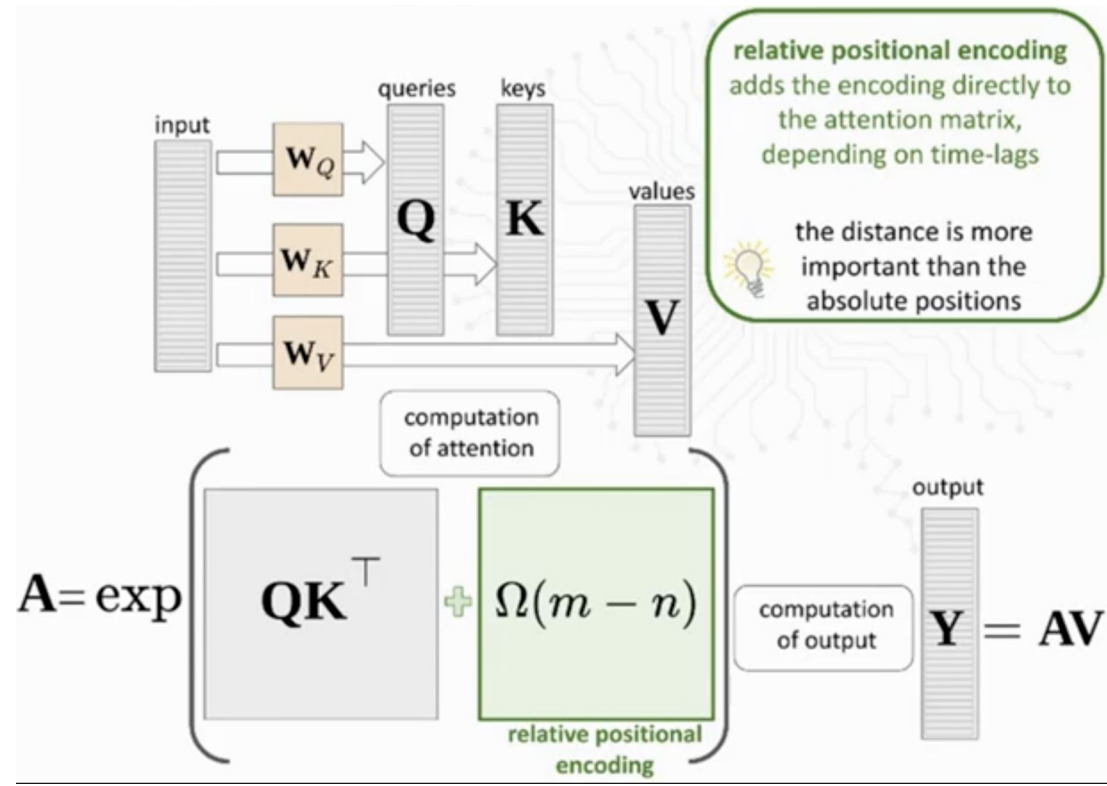
$$\psi'(m) = \sum_{i=0}^{\frac{D}{2}} -w_i \sin(w_i m)$$

- For Transformer, $w_i = (\frac{1}{10000})^{2i/D}$, so $\psi(m)$ is only monotonic when m is approximately in the range of 0 to 50.

⇒ Position encoding is not sensitive to distances greater than 50.

⇒ To address this, learnable positional encoding can be used. In this case, the frequencies are learned based on the training data. Learnable positional encoding is **positional embedding**.

Relative Positional Embedding



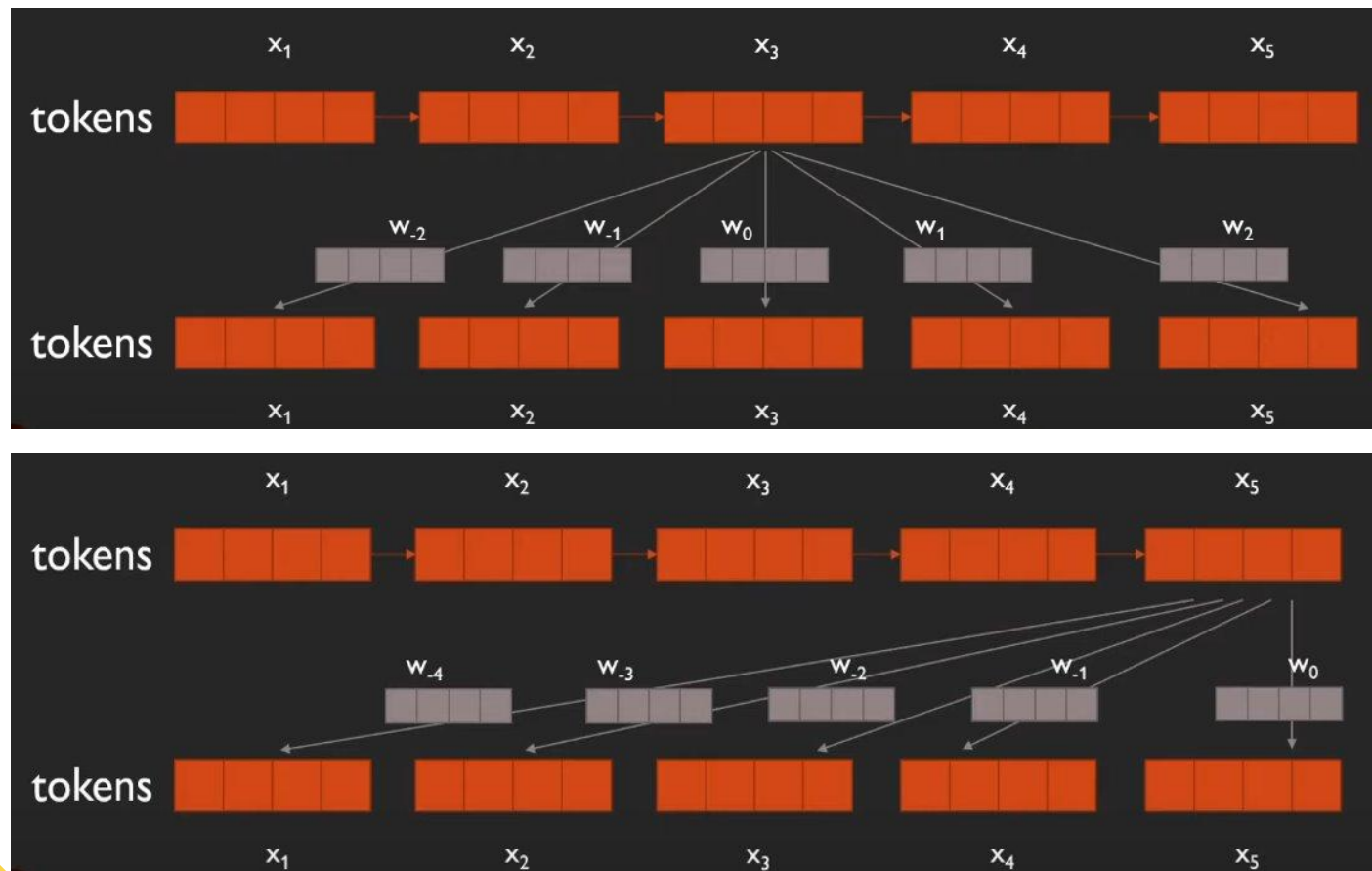
Motivation

The issue with absolute PEs:

Position	1	2	3	4	5	6
	I	walk my	dog	every		day

Position	1	2	3	4	5	6
	every	day	I	walk my	dog	

Relative PEs



Absolute-Relative PEs

Let the position space be $\mathcal{P}$ and the input embedding space be $\mathcal{X}$.

The task of the positional embedding function is to learn the mapping $F: \mathcal{P} \rightarrow \mathcal{X}$.

- Absolute PEs map each position in the sentence to an element in the position space.
- Relative PEs map the distance between two positions in the sentence to an element in the position space.

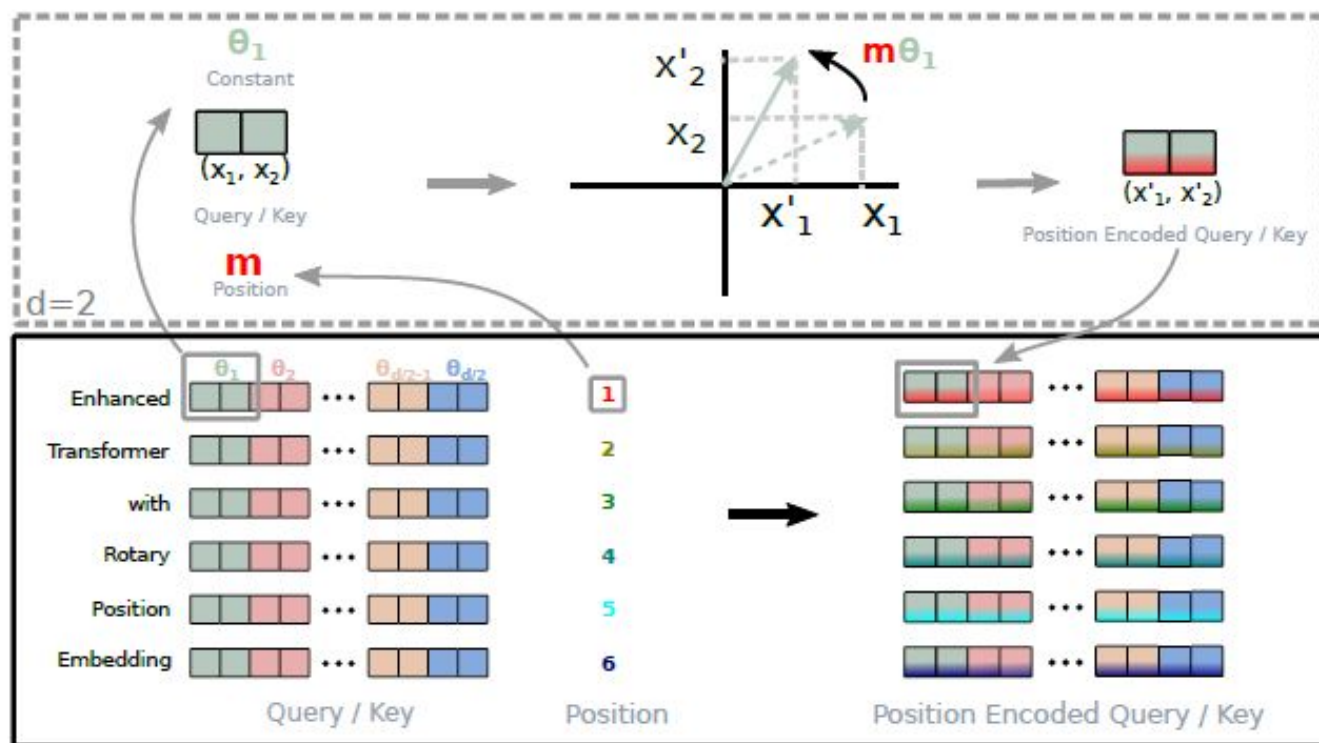
Absolute-Relative PEs

The self-attention mechanism will be modified depending on the type of positional embedding used:

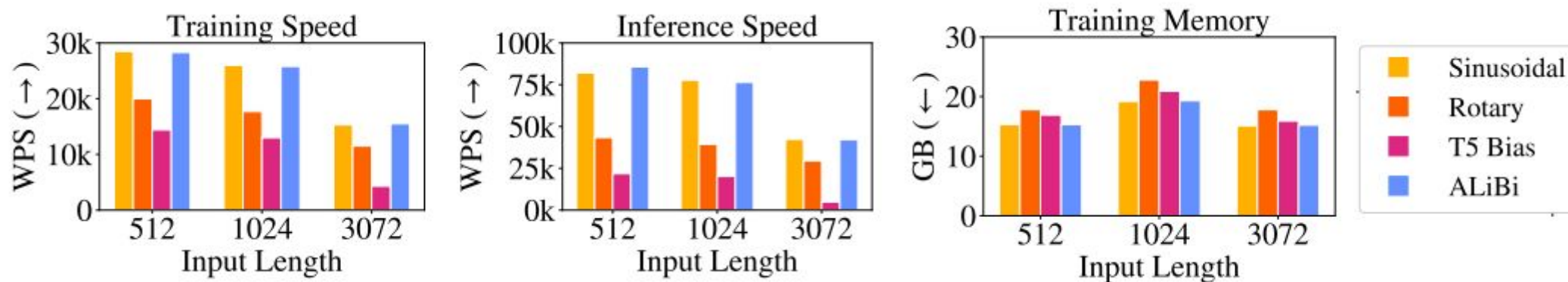
Absolute PEs:
$$\begin{bmatrix} Q_x \\ K_x \\ V_x \end{bmatrix} = (WE_x + P_x) \odot \begin{bmatrix} W^Q \\ W^K \\ W^V \end{bmatrix}$$

Relative PEs:
$$\begin{bmatrix} Q_x \\ K_x \\ V_x \end{bmatrix} = WE_x \odot \begin{bmatrix} W^Q \\ W^K \\ W^V \end{bmatrix} + \begin{bmatrix} 0 \\ P_{x-y} \\ P_{x-y} \end{bmatrix}$$

Rotary Positional Embedding

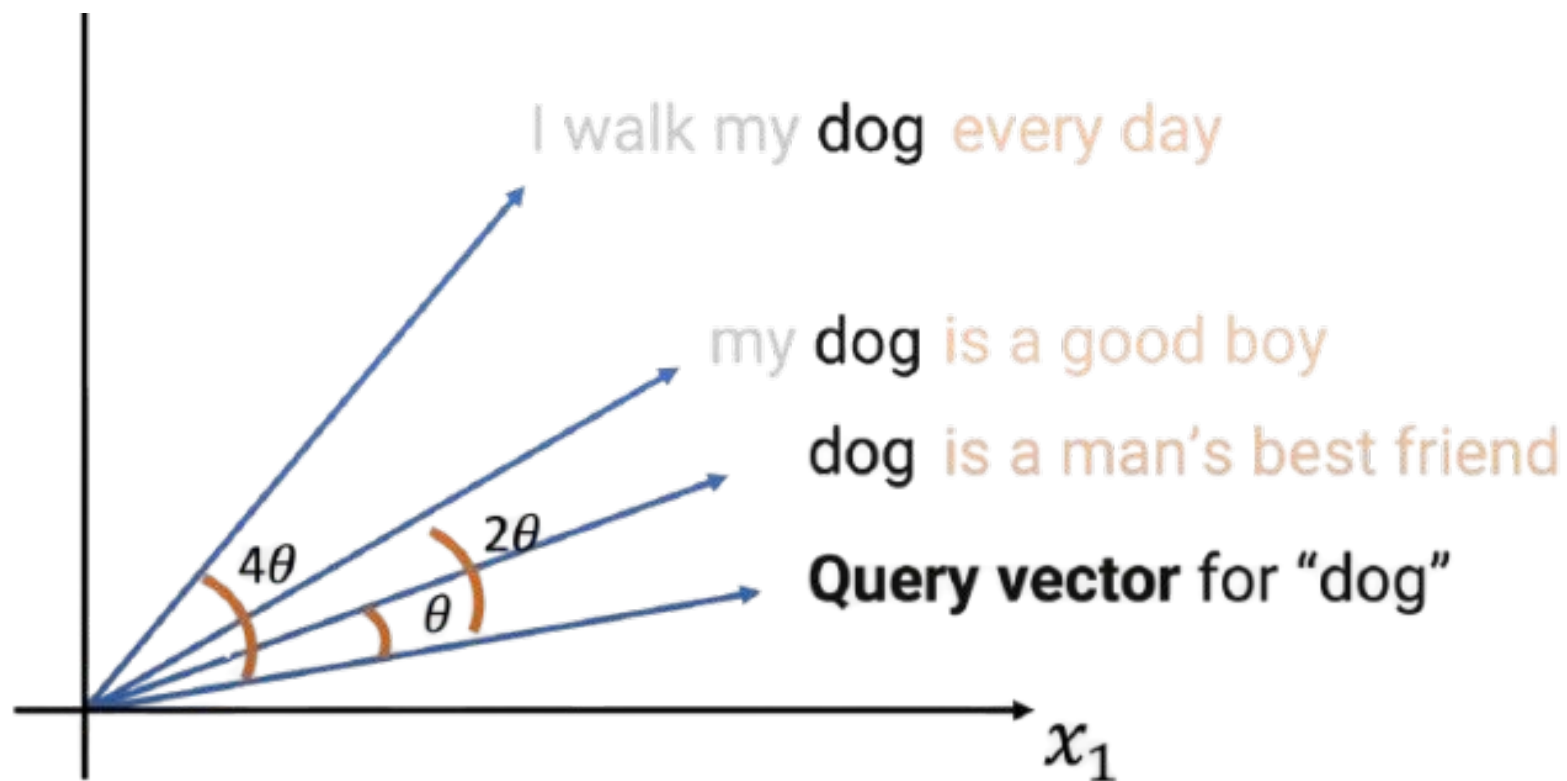


Motivation

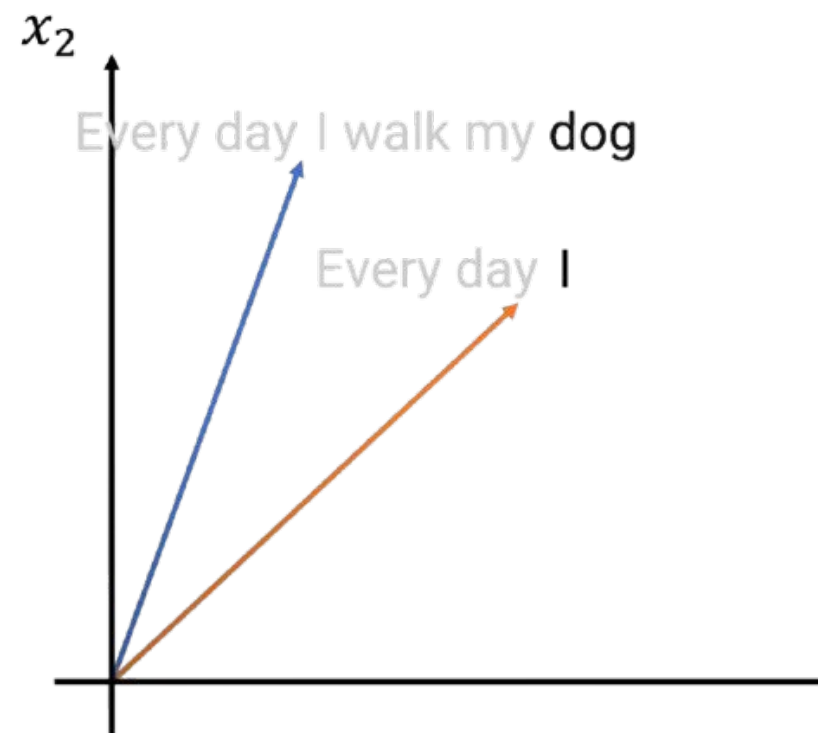
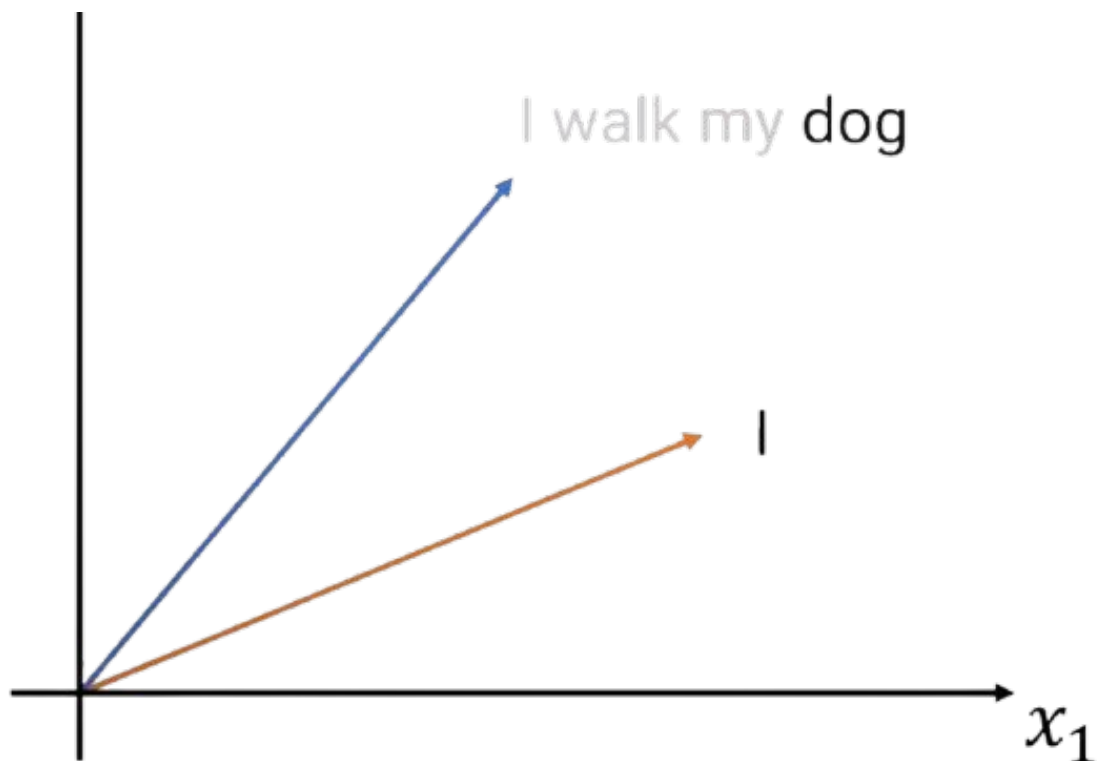


- Extra step in self-attention layer.
- Changes in every step -> difficult for KV cache.

Rotary Positional Embedding



Rotary Positional Embedding



Rotary Positional Embedding

- Query vector for “dog”:

$$q_{\text{dog}} = \begin{bmatrix} \cos(4\theta) & -\sin(4\theta) \\ \sin(4\theta) & \cos(4\theta) \end{bmatrix} W_q x_{\text{dog}}$$

- Key vector for “I”:

$$k_I = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} W_k x_I$$

- Therefore:

$$\alpha_{\text{dog}, I} = x_I^T W_k^T \begin{bmatrix} \cos(3\theta) & -\sin(3\theta) \\ \sin(3\theta) & \cos(3\theta) \end{bmatrix} W_q x_{\text{dog}}$$

Rotary Positional Embedding

- Query vector for “dog”:

$$q_{\text{dog}} = \begin{bmatrix} \cos(6\theta) & -\sin(6\theta) \\ \sin(6\theta) & \cos(6\theta) \end{bmatrix} W_q x_{\text{dog}}$$

- Key vector for “I”:

$$k_I = \begin{bmatrix} \cos(3\theta) & -\sin(3\theta) \\ \sin(3\theta) & \cos(3\theta) \end{bmatrix} W_k x_I$$

- Therefore:

$$\alpha_{\text{dog}, I} = x_I^T W_k^T \begin{bmatrix} \cos(3\theta) & -\sin(3\theta) \\ \sin(3\theta) & \cos(3\theta) \end{bmatrix} W_q x_{\text{dog}}$$

Rotary Positional Embedding

A 2D Case:

$$R_{\Theta}^m = \begin{bmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{bmatrix}$$

- Query vector for the token at position m :

$$q_m = R_{\Theta}^m W_q x_m$$

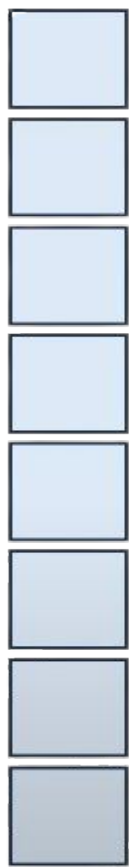
- Key vector for the token at position n :

$$k_n = R_{\Theta}^n W_k x_n$$

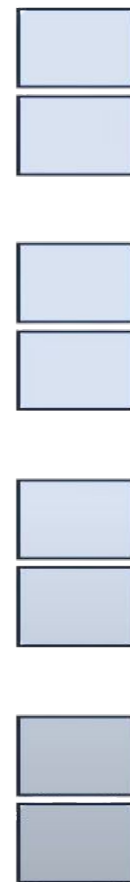
- Therefore:

$$\alpha_{m,n} = x_n^T W_k^T (R_{\Theta}^n)^T R_{\Theta}^m W_q x_m = x_n^T W_k^T R_{\Theta}^{-n} R_{\Theta}^m W_q x_m = x_n^T W_k^T R_{\Theta}^{m-n} W_q x_m$$

Rotary Positional Embedding



Devide the d -dimension space into $d/2$ sub-spaces



Rotary Positional Embedding

General form:

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

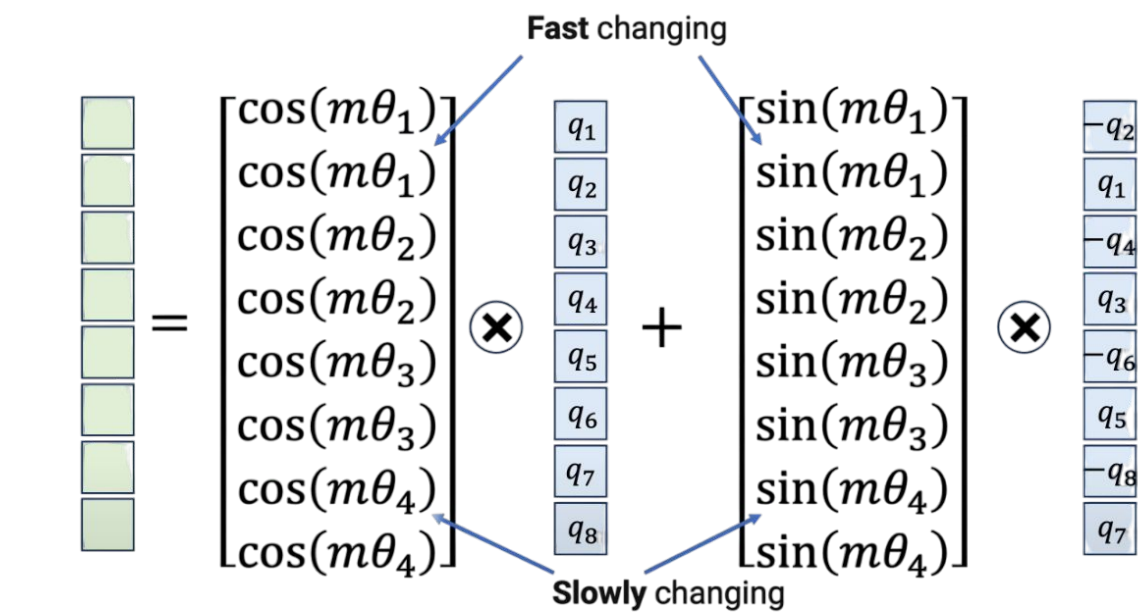
Rotary Positional Embedding

General form:

$$R_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \ddots & & & \\ 0 & 0 & & \ddots & & \\ \vdots & \vdots & & & \ddots & \\ 0 & 0 & & & & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & & & & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

$$R_{\Theta, m}^d \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix}$$

Rotary Positional Embedding



$f_q(\mathbf{x}_m, m)$

$$\theta_i = N^{-2i/d}$$

$$N = 10,000$$

High Frequencies: Positional attention
Low Frequencies: Semantic attention

Rotary Positional Embedding

$$\alpha = \frac{L_{train}}{L_{target}}$$

Example: $\alpha = \frac{4K}{20K} = \frac{1}{5}$

What if $m > L_{train}$?

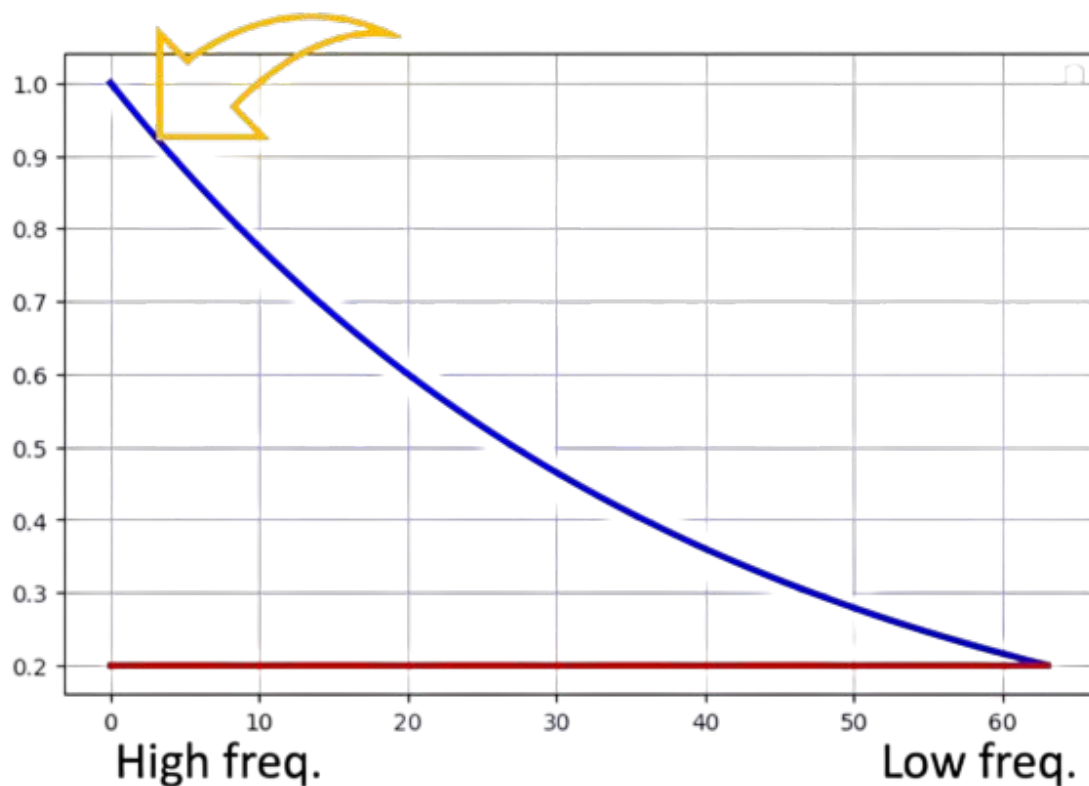
$$\begin{bmatrix} \square \\ \square \end{bmatrix} = \begin{bmatrix} \cos(m\alpha\theta_1) & -\sin(m\alpha\theta_1) \\ \sin(m\alpha\theta_1) & \cos(m\alpha\theta_1) \end{bmatrix} \begin{bmatrix} \square \\ \square \end{bmatrix}$$

$$\begin{bmatrix} \square \\ \square \end{bmatrix} = \begin{bmatrix} \cos(m\alpha\theta_2) & -\sin(m\alpha\theta_2) \\ \sin(m\alpha\theta_2) & \cos(m\alpha\theta_2) \end{bmatrix} \begin{bmatrix} \square \\ \square \end{bmatrix}$$

$$\begin{bmatrix} \square \\ \square \end{bmatrix} = \begin{bmatrix} \cos(m\alpha\theta_3) & -\sin(m\alpha\theta_3) \\ \sin(m\alpha\theta_3) & \cos(m\alpha\theta_3) \end{bmatrix} \begin{bmatrix} \square \\ \square \end{bmatrix}$$

$$\begin{bmatrix} \square \\ \square \end{bmatrix} = \begin{bmatrix} \cos(m\alpha\theta_4) & -\sin(m\alpha\theta_4) \\ \sin(m\alpha\theta_4) & \cos(m\alpha\theta_4) \end{bmatrix} \begin{bmatrix} \square \\ \square \end{bmatrix}$$

Rotary Positional Embedding



Linear

$$\theta'_i = \alpha \theta_i \quad \alpha = \frac{4K}{20K} = \frac{1}{5}$$

Neural Tangent Kernel — (NTK)

$$\theta'_i = \alpha \theta_i \quad a_i = 5^{-\frac{2i}{d_k-2}}$$

There are two other types of RoPE: YaRN and CoPE.



Thank you for your attention

Quoc Le

llongquoc1999@gmail.com